

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Agricultural Economics: Demand & Supply

Introduction

- Agricultural economics applies economic principles to farming. Demand and supply are its foundation: the demand for farm products and the supply of farm products determine farm prices and incomes.

Demand for farm products

- The demand for food is price inelastic in the short run (people must eat); income elasticity is positive for protein foods (meat, fish, milk) — incomes rise and consumption shifts from staples to livestock products; the demand is also affected by population, tastes (festive seasons), urbanisation and the price of substitutes.

Supply of farm products

- Supply is inelastic in the short run (the crop cannot be hurried — the gestation period) and elastic in the long run (farmers respond to prices by planting more); supply depends on weather, inputs, technology, storage and the producer's price expectations.

The farm price determination

- Seasonal price fluctuations (cheap at harvest, dear in the lean season — the reason for storage and the stabilisation schemes); the cobweb cycle (high prices → more planting → glut → low prices → less planting); government influence (minimum prices, marketing boards, subsidies).

CLASS EVALUATION

1. Why is the demand for staple food inelastic?
2. Explain short-run vs long-run supply in agriculture.
3. What is the cobweb cycle?
4. Give two causes of seasonal price changes.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Farming as a Business (Farm Management)

Introduction

- Farm management is the science of organising and operating a farm for profit — land, labour, capital and management (the four factors) combined efficiently. The modern view: farming is a business, not a way of life.

Functions of management

- **Planning** — selecting enterprises, budgeting (the farm plan and the budget); **organising** — acquiring and arranging resources; **directing** — supervising labour and operations; **controlling** — records, accounting, evaluating results; **decision-making** — choosing enterprises, inputs and markets; and **risk management** — insurance, diversification, storage.

Costing and profitability

- Fixed costs (rent, interest, depreciation) and variable costs (seed, feed, fertiliser, casual labour); total cost = fixed + variable; revenue = output × price; profit = revenue – total cost; the break-even point; efficiency measures: output per hectare, output per labour unit, the cost per unit of produce.

Business planning tools

- The enterprise budget, cash-flow forecast, the record-keeping system (the farm diary), and the analysis of alternatives (which enterprise gives the best return on capital?).

CLASS EVALUATION

1. Define farm management.
2. State the five functions of management.
3. How is farm profit calculated?
4. What is the break-even point and why does it matter to the farmer?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Agricultural Policies & Finance

Introduction

- Government policies shape agriculture: the incentives, institutions and funds that help or hinder the farmer. Agricultural finance is the provision of credit to agriculture — the capital that buys inputs and machinery.

Policies and programmes

- The national agricultural policies (food security, self-sufficiency); commodity programmes (the defunct marketing boards; the current anchor borrowers' scheme — the CBN's intervention for rice, wheat and other crops); the National Fadama programme (irrigation and rural development); the Growth Enhancement Support Scheme (fertiliser, improved seed); the National Agricultural Land Development Authority (NALDA) — land development and mechanisation; and the tariff protection of local produce.

Sources of farm credit

- **Formal** — commercial banks, development banks (the Bank of Agriculture), microfinance banks, cooperative societies and the government's intervention funds (the Anchor Borrowers' Programme — low-interest credit to smallholder farmers); **informal** — money lenders (dear), traders, friends and the family (easy but inefficient). Problems of credit: collateral, high interest, poor record keeping and the default risk — hence group lending and the cooperative model.

CLASS EVALUATION

1. State two agricultural policies and their objectives.
2. What is the Anchor Borrowers' Programme?
3. Name four sources of farm credit.
4. Why do banks hesitate to lend to smallholders?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Agricultural Cooperatives

Introduction

- A cooperative society is a voluntary association of persons who pool resources for mutual benefit. Agricultural cooperatives are the farmer's answer to the problems of small scale: they buy inputs in bulk, market produce together, give credit and provide services no single farmer can afford.

Types

- **Producer/marketing cooperatives** — assemble and sell members' produce (better prices, grading); **credit and thrift cooperatives** — savings and loans to members; **consumer cooperatives** — bulk buying of inputs and consumer goods (fertiliser, seed, machinery hire); **multipurpose cooperatives** — the commonest, combining functions; and **specialised cooperatives** — by crop (cocoa, rice, cassava) or animal.

Principles and organisation

- Voluntary and open membership; democratic control (one member, one vote); limited interest on capital; the equitable distribution of surplus (patronage refunds); education; and cooperation between societies. Organisation: the general meeting (the highest body), the executive committee, the treasurer and the secretariat.

Importance and problems

- Importance: economies of scale, bargaining power, credit for members, training and extension, the diffusion of technology and the group marketing of produce. Problems: poor management and dishonesty, inadequate capital, the lack of member loyalty, political interference and the difficulty of auditing small societies.

CLASS EVALUATION

1. Define a cooperative.
2. State three types of agricultural cooperatives.
3. List four principles of cooperation.
4. Mention three problems of cooperatives.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Agricultural Extension & Research (Revision)

Introduction

- This week revises the service that links agricultural research to the farm — extension — and the research institutions that produce the improved technologies: seed, feed, vaccines and practices.

The extension system

- The Agricultural Development Programme (ADP) structure: the zonal and block extension agents (the training-and-visit system) working with village extension workers and farmer groups; the Federal Agricultural Extension and Research Liaison Services (NAERLS, Zaria); the National Research Institutes (IITA — cassava, maize; NCRI — rice; NRCRI — root crops; NIFOR — oil palm; NRI — rubber; NVRI — veterinary research, Vom).

Transfer of technology

- Demonstration plots, on-farm trials, field days, farmer field schools, radio and mobile-phone advisories, and the mass media. The National Agricultural Technology and Information System (NATIS) coordinates the transfer. Successful extension has the farmer as the partner, not the pupil.

CLASS EVALUATION

1. What is agricultural extension?
2. Name three research institutes and one crop each.
3. What is a farmer field school?
4. Why is extension the link between research and the farmer?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Agriculture & National Development

Introduction

- Agriculture is the backbone of the Nigerian economy: it feeds the nation, employs the people, earns foreign exchange and supplies industry with raw materials. This final week surveys its contribution and its food-security role.

Contribution to the economy

- Employment — about 35% of the labour force (the largest sector, though urban services are growing); GDP — about 20–25%; foreign exchange — cocoa, sesame seeds, cashew nuts, shea butter, gum arabic, ginger and rubber; raw materials — cotton for textiles, palm oil for industry; and the market for domestic industry (food processing, tractors, agrochemicals).

Food security

- Food security means every person having physical and economic access to sufficient, safe and nutritious food at all times. The four pillars: **availability** (production), **access** (affordability and markets), **utilisation** (nutrition and safety) and **stability** (all year round). Nigeria's challenges: the imports of wheat, rice, sugar and fish; the post-harvest losses; insecurity in the food belts; and the high cost of inputs.

Government strategies

- The agricultural promotion policies, the Green Imperative Programme (mechanisation and the Brazilian partnership), the national strategic food reserves, the CBN's Anchor Borrowers' Programme (credit for thousands of smallholder farmers in rice, wheat and other crops), the National Agricultural Land Development Authority (NALDA), the Fadama projects (irrigation and rural infrastructure), and the promotion of the 'Made in Nigeria' food campaign.

CLASS EVALUATION

1. State two ways agriculture contributes to the economy.
2. Define food security.
3. Name the four pillars of food security.
4. State two strategies that improve Nigerian food security.
5. Why does Nigeria still import rice?

